

A03 Self-reflection essay

I intentionally started by testing very small integers, mostly from 1 to 4, and playing with different combinations of the base and the number of iterations. At first, I noticed that when the iteration count was small, GPT-4.1 nano could usually give the correct answer, even when the base—the number before the comma—was relatively large, such as 3 or 4. However, as I increased the number of iterations, the model seemed more likely to make mistakes. Sometimes it appeared to stop one iteration too early. This was interesting because the Python function always produced the correct result, while GPT's answers were not consistent.

After attending three classes and reading the first few chapters of *Introduction to Generative AI*, I started to have some idea on why this happened. I “blame” these mistakes on how a large language model works. An LLM does not really calculate in the same way as a calculator or a Python function. Instead, it predicts the next token based on which token is most likely to come next (highest probabilities). Because of this, it may recognize the pattern of repeated squaring without actually performing every calculation accurately. As the calculation becomes longer, there are more opportunities for it to lose track of the number of iterations.

I also set the temperature to 0.2 rather than 0.0. This allows a small amount of randomness in the model's token selection, which may make its answers slightly less precise. Overall, this experiment helped me see the difference between generating a likely answer and actually calculating a correct answer.